

UME410: Mechatronic Systems

L	T	P	Cr
3	0	2	4.0

Course Objective: The course imparts interdisciplinary knowledge to study modern products like household appliances, digital cameras, mobiles etc., which falls under the mechatronics domain. The aim of this course to make a bridge between mechanical, electronics, instrumentation, computer and controls field.

Syllabus

Introduction: Evolution of mechatronics, integrated mixed systems, integration of mechanical engineering, electronics and control engineering and computer science, design process, measurement system, control system, basic elements of open-loop and closed-loop control system, block diagram representation of mechatronics system, sequential controllers.

SISO Control Systems: Performance specifications, transfer functions, block diagram reduction techniques, signal flow graphs, sensitivity analysis, frequency response, stability, controller types and their design using frequency domain and Laplace domain method, PID control, feedback control.

MIMO Control Systems: Effect of pole-zero cancellations, frequency vs time-domain control, state-space representation, linear transformations in state-space representation, system characteristics from state-space representation.

Dynamic Systems Modeling: Equations of motion of mechanical, hydraulic, thermal, electric and pneumatic systems, modeling of motors and generators, solving the dynamic model in MATLAB environment.

Sensors (Flip Learning Component): Performance terminology, static and dynamic characteristics, displacement, position and proximity sensors, velocity and motion sensors, stress, strain and force measurements using strain gauges, force, fluid pressure, liquid flow and liquid level sensors, light sensors, temperature sensors.

Signal Conditioning and Digital Signals: Basic conditioning process, operational amplifiers, digital signal, AD and DA conversion, Shannon's sampling theorem,

Electronic Elements in Mechatronic System: Introduction to microprocessors and microcontrollers.

Laboratory Work

Experiments based on Lego kit, Tetrix kit, microcontroller based kits, different sensors, interfacing with PC, modeling and control through servo motors, data acquisition related experiments like Quanser Qube or SRV-02 workstation in MATLAB/Simulink/Labview environment.